# TEAS 7 Science Review Sheets

Review of the four TEAS 7 Science sub-areas: anatomy & physiology, biology, chemistry, scientific reasoning

MindShelf

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## The Science Section at a Glance

| Sub-area (ATI TEAS 7 content outline) | Scored items |
|---|---|
| **Human anatomy & physiology**: Body orientation; respiratory, cardiovascular, digestive, nervous, muscular, skeletal, reproductive, integumentary, endocrine, urinary and immune systems | 18 |
| **Biology**: Cells; DNA, RNA and proteins; Mendelian inheritance; macromolecules; microorganisms and disease | 9 |
| **Chemistry**: Atomic structure; properties and changes of matter; reactions and reaction conditions; solutions; acids and bases | 8 |
| **Scientific reasoning**: Measurements and tools; logic and evidence; relationships between variables; the scientific method | 9 |

44 scored items plus 6 unscored pretest items (50 in all) in 60 min. Pretest items are mixed in with the scored ones and do not count, so give every question full effort. These sheets review core facts for each sub-area; they are not a complete course and contain no actual test questions.

## Human Anatomy & Physiology

Largest sub-area: 18 scored items.

### Orientation

**Anatomical position:** standing upright, facing forward, arms at the sides, palms forward, feet parallel. **Supine** = lying face up; **prone** = face down. Left and right always mean the *patient's* left and right.

| Term | Opposite | Meaning |
|---|---|---|
| **Superior (cranial)** | Inferior (caudal) | Toward the head / toward the feet |
| **Anterior (ventral)** | Posterior (dorsal) | Front / back of the body |
| **Medial** | Lateral | Toward / away from the midline |
| **Proximal** | Distal | Closer to / farther from the trunk or a limb's attachment |
| **Superficial** | Deep | Toward / away from the body surface |

| Plane | Divides the body into |
|---|---|
| **Sagittal (midsagittal = equal halves)** | Right and left parts |
| **Frontal (coronal)** | Anterior and posterior parts |


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| Plane | Divides the body into |
|---|---|
| **Transverse (horizontal)** | Superior and inferior parts |

**Cavities:** dorsal = cranial (brain) + vertebral (spinal cord); ventral = thoracic (pleural cavities, pericardial cavity, mediastinum) + abdominopelvic, separated by the diaphragm.

**Levels of organization:** chemical → cell → tissue → organ → organ system → organism. Tissue types: epithelial, connective, muscle, nervous.

**Homeostasis:** negative feedback reverses a change (body temperature near 37 °C / 98.6 °F, blood glucose, blood pH 7.35–7.45); positive feedback amplifies it until an end point (oxytocin in labor, blood clotting).

## Cardiovascular System

| | |
|---|---|
| **Right atrium** | Receives deoxygenated blood from the superior and inferior venae cavae |
| **Right ventricle** | Pumps blood through the pulmonary valve into the pulmonary trunk (pulmonary circuit) |
| **Left atrium** | Receives oxygenated blood from the pulmonary veins |
| **Left ventricle** | Thickest wall; pumps blood through the aortic valve into the aorta (systemic circuit) |
| **Valves** | Tricuspid (right) and mitral or bicuspid (left); semilunar valves (pulmonary and aortic) guard the outflow; all prevent backflow |
| **Conduction** | SA node (pacemaker) → AV node → AV bundle (bundle of His) → bundle branches → Purkinje fibers |
| **Vessels** | Arteries carry blood away from the heart; capillaries (one cell thick) exchange gases and nutrients; veins return blood, with valves in the limbs |
| **Blood** | Erythrocytes carry $O_2$ on hemoglobin; leukocytes defend; platelets start clotting; plasma carries solutes |

4 chambers. Blood path: venae cavae → right atrium → tricuspid → right ventricle → pulmonary arteries → lungs → pulmonary veins → left atrium → mitral → left ventricle → aorta → body.

**Cardiac output**

$$CO = HR \times SV$$

*E.g.* HR 70/min, SV 70 mL: CO = 70 × 70 mL/min = 4.9 L/min.

| Type | RBC antigens | Plasma antibodies | Can receive red cells from (ABO only) |
|---|---|---|---|
| **A** | A | anti-B | A, O |
| **B** | B | anti-A | B, O |
| **AB** | A and B | none | A, B, AB, O |
| **O** | none | anti-A and anti-B | O only |

**Rh:** Rh-positive cells carry the D antigen. An Rh-negative person makes anti-D only after exposure (transfusion, pregnancy with an Rh-positive fetus), which can cause hemolytic disease of the newborn in a later pregnancy. **Universal red-cell donor:** O−; **universal recipient:** AB+. An Rh− person should receive Rh− red cells.

## Respiratory System

| | |
|---|---|
| **Nasal cavity** | Warms, humidifies and filters inhaled air |
| **Pharynx** | Throat: shared passage for air and food |
| **Larynx** | Airway only: holds the vocal cords; the epiglottis covers it during swallowing |
| **Trachea and bronchi** | Cartilage rings keep the trachea open; bronchi branch into bronchioles (conducting zone) |
| **Alveoli** | Gas exchange by diffusion across a thin respiratory membrane; surfactant lowers surface tension |
| **Diaphragm** | Contracts and flattens on inspiration: chest volume up, pressure down, air flows in |
| **Brainstem centers** | Medulla and pons set the breathing rhythm; rising $CO_2$ (falling pH) is the main drive to breathe |

Right lung 3 lobes, left lung 2 lobes (room for the heart). $O_2$ and $CO_2$ move by diffusion, from higher to lower partial pressure.

## Digestive System

| | |
|---|---|
| **Mouth** | Chewing (mechanical digestion); salivary amylase starts starch digestion |
| **Esophagus** | Moves the bolus to the stomach by peristalsis |
| **Stomach** | HCl and pepsin begin protein digestion; churning makes chyme |
| **Small intestine** | Duodenum, jejunum, ileum: most chemical digestion and absorption; villi and microvilli enlarge the surface |
| **Liver and gallbladder** | Liver makes bile (emulsifies fats); gallbladder stores and concentrates it |

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| | |
|---|---|
| **Pancreas (exocrine)** | Releases amylase, lipase, proteases and bicarbonate into the duodenum |
| **Large intestine** | Absorbs water and electrolytes; gut bacteria make vitamin K; forms feces |

Order: mouth → pharynx → esophagus → stomach → small intestine → large intestine → rectum → anus. Liver, gallbladder and pancreas are accessory organs: food never passes through them.

## Nervous System

| | |
|---|---|
| **CNS** | Brain and spinal cord: integrate information |
| **Neuron** | Dendrites receive, the cell body integrates, the axon conducts; myelin (Schwann cells in the PNS, oligodendrocytes in the CNS) speeds conduction |
| **Cerebrum** | Lobes: frontal (motor, planning, personality), parietal (touch, body sense), temporal (hearing, memory), occipital (vision) |
| **Cerebellum** | Coordination, balance, posture |
| **Brainstem** | Midbrain, pons, medulla oblongata: breathing, heart rate, blood pressure |
| **Autonomic NS** | Sympathetic: fight or flight (heart rate up, pupils dilate, airways dilate, digestion slows); parasympathetic: rest and digest |
| **Action potential** | $Na^+$ rushes in (depolarization), $K^+$ flows out (repolarization); neurotransmitters such as acetylcholine cross the synapse |

PNS: 12 pairs of cranial nerves, 31 pairs of spinal nerves; afferent (sensory) toward the CNS, efferent (motor) away from it.

## Muscular & Skeletal Systems

| | |
|---|---|
| **Skeletal muscle** | Voluntary, striated, multinucleated fibers |
| **Cardiac muscle** | Involuntary, striated, branched cells joined by intercalated discs |
| **Smooth muscle** | Involuntary, not striated; walls of hollow organs and blood vessels |
| **Sarcomere** | Contraction unit: actin (thin) and myosin (thick) filaments slide past each other; needs $Ca^{2+}$ and ATP |
| **Bone functions** | Support, protection, movement, mineral ($Ca^{2+}$) storage, blood cell formation in red marrow |
| **Bone cells** | Osteoblasts build bone, osteoclasts break it down, osteocytes maintain it |
| **Connections** | Tendons join muscle to bone; ligaments join bone to bone; synovial joints move freely |

Adult skeleton: 206 bones. Axial = skull, vertebral column, rib cage; appendicular = limbs and their girdles. Vertebrae: 7 cervical, 12 thoracic, 5 lumbar, then sacrum and coccyx.

## Endocrine System

| | |
|---|---|
| **Hypothalamus** | Releasing and inhibiting hormones control the anterior pituitary; makes ADH and oxytocin |
| **Anterior pituitary** | GH, TSH, ACTH, FSH, LH, prolactin |
| **Posterior pituitary** | Releases ADH (water reabsorption) and oxytocin (uterine contraction, milk ejection) |
| **Thyroid** | $T_4$ and $T_3$ raise metabolic rate; calcitonin lowers blood $Ca^{2+}$ |
| **Parathyroids** | PTH raises blood $Ca^{2+}$ |
| **Adrenal cortex** | Cortisol (stress, raises glucose); aldosterone ($Na^+$ and water retention) |
| **Adrenal medulla** | Epinephrine and norepinephrine (fight or flight) |
| **Pancreatic islets** | Insulin from beta cells lowers blood glucose; glucagon from alpha cells raises it |
| **Pineal gland** | Melatonin (sleep–wake rhythm) |

Most hormone levels are held steady by negative feedback (e.g., high $T_4$ suppresses TSH). **Endocrine** glands are ductless and release hormones into the blood; **exocrine** glands secrete through ducts (sweat, salivary, the enzyme-secreting part of the pancreas).

## Urinary System

| | |
|---|---|
| **Kidneys** | Filter blood; regulate water, electrolytes, pH and blood pressure; release renin and erythropoietin (EPO) |
| **Nephron** | Glomerulus and glomerular (Bowman's) capsule filter → proximal tubule (most reabsorption) → nephron loop (loop of Henle) → distal tubule → collecting duct |
| **Hormonal control** | ADH increases water reabsorption in collecting ducts; aldosterone increases $Na^+$ reabsorption |
| **Urine path** | Ureters (peristalsis) → urinary bladder (storage) → urethra |

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## Reproductive System

| | |
|---|---|
| **Testes** | Sperm production in seminiferous tubules; testosterone from interstitial cells |
| **Male ducts and glands** | Epididymis (maturation, storage) → ductus (vas) deferens → urethra; seminal vesicles, prostate and bulbourethral glands add semen fluid |
| **Ovaries** | Oocytes; estrogen and progesterone |
| **Uterine tubes and uterus** | Fertilization usually occurs in the uterine (fallopian) tube; implantation in the endometrium, which sheds at menstruation |
| **Hormones** | FSH and LH from the anterior pituitary drive gamete production; the LH surge triggers ovulation |

## Integumentary System

| | |
|---|---|
| **Epidermis** | Keratinized stratified squamous epithelium, no blood vessels; melanocytes make melanin (UV protection) |
| **Dermis** | Connective tissue with blood vessels, nerves, hair follicles, sweat and sebaceous glands |
| **Hypodermis** | Subcutaneous fat: insulation, energy storage, anchoring |
| **Functions** | Barrier, thermoregulation (sweat, blood flow), sensation, vitamin D synthesis |

## Immune & Lymphatic Systems

| | |
|---|---|
| **Innate defenses** | Nonspecific: skin and mucous membranes, phagocytes (neutrophils, macrophages), NK cells, inflammation, fever |
| **B cells** | Humoral immunity: become plasma cells that secrete antibodies; mature in red bone marrow |
| **T cells** | Cell-mediated immunity: helper T cells coordinate, cytotoxic T cells kill infected cells; mature in the thymus |
| **Lymphatic organs** | Lymph nodes filter lymph; spleen filters blood and removes old red cells; lymphatic vessels return fluid to the blood |
| **Immunity types** | Active (infection or vaccine, makes memory cells) vs passive (antibodies received, e.g., across the placenta; short-lived) |

# Biology

9 scored items.

## Cells & Organelles

| Structure | Function |
|---|---|
| **Nucleus** | Holds the DNA; the nucleolus assembles ribosome subunits |
| **Ribosome** | Builds proteins (free in cytosol or bound to rough ER) |
| **Rough ER** | Ribosome-studded; makes proteins for membranes and export |
| **Smooth ER** | Makes lipids, detoxifies, stores $Ca^{2+}$ |
| **Golgi apparatus** | Modifies, sorts and packages proteins into vesicles |
| **Mitochondrion** | Aerobic cellular respiration → ATP; has its own DNA |
| **Lysosome** | Digestive enzymes break down waste and worn-out parts (animal cells) |
| **Cell membrane** | Phospholipid bilayer with proteins (fluid mosaic), selectively permeable |
| **Cytoskeleton** | Microtubules and microfilaments: shape, transport, movement |
| **Centrioles** | Organize the spindle in animal cells |
| **Plant cells (not animal cells)** | Cell wall (cellulose), chloroplasts (photosynthesis), large central vacuole |

**Prokaryotes** (bacteria, archaea): no nucleus, no membrane-bound organelles; DNA in a nucleoid; ribosomes present. **Eukaryotes** (animals, plants, fungi, protists): nucleus and organelles.

**Membrane transport.** Passive, no ATP, down the gradient: diffusion, facilitated diffusion, osmosis (water moves toward the higher solute concentration). Active, uses ATP, against the gradient: e.g., the $Na^+/K^+$ pump, endocytosis, exocytosis.

| Cell placed in | Result |
|---|---|
| **Hypertonic** solution | Water leaves; the cell shrinks (crenates) |
| **Hypotonic** solution | Water enters; the cell swells, may burst (lyse) |
| **Isotonic** solution | No net water movement |

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**Cellular respiration and photosynthesis**

**Aerobic respiration:** $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + ATP$. Glycolysis (cytoplasm, no $O_2$ needed) → citric acid (Krebs) cycle (mitochondrial matrix) → electron transport chain (inner mitochondrial membrane; $O_2$ is the final electron acceptor; makes most of the ATP). Without $O_2$, cells rely on fermentation (lactic acid in human muscle).

**Photosynthesis** (chloroplasts): $6\ CO_2 + 6\ H_2O + light \rightarrow C_6H_{12}O_6 + 6\ O_2$. Its products are the reactants of respiration.

## Cell Division

**Cell cycle:** interphase ($G_1$ growth, S DNA replication, $G_2$ preparation) → mitosis: prophase, metaphase (chromosomes line up), anaphase (sister chromatids separate), telophase (PMAT) → cytokinesis.

| | Mitosis | Meiosis |
|---|---|---|
| **Purpose** | Growth, repair, asexual reproduction | Gametes (eggs and sperm) |
| **Divisions** | One | Two (meiosis I and II) |
| **Daughter cells** | Two genetically identical cells | Four genetically different cells |
| **Chromosome number** | Diploid (2n) kept | Haploid (n): halved |
| **Variation** | None (barring mutation) | Crossing over (prophase I) and independent assortment |

Humans: 2n = 46 (23 pairs); a gamete has $n = 46/2 = 23$.

## DNA, RNA & Protein Synthesis

| | DNA | RNA |
|---|---|---|
| **Sugar** | Deoxyribose | Ribose |
| **Bases** | A, T, G, C | A, U, G, C |
| **Strands** | Double helix | Usually single |
| **Role** | Stores genetic information | Carries it (mRNA), brings amino acids (tRNA), forms ribosomes (rRNA) |

Base pairing: A–T and G–C in DNA; A–U in RNA. DNA replication is semi-conservative (each new helix keeps one old strand).

**Central dogma**
DNA → **transcription** (nucleus, RNA polymerase) → mRNA → **translation** (ribosome, tRNA brings amino acids) → protein

A **codon** is 3 bases of mRNA; the tRNA **anticodon** pairs with it. $4^3 = 64$ codons code for 20 amino acids plus 3 stop codons (UAA, UAG, UGA): the code is redundant. AUG = start (methionine). *E.g.* template DNA TAC CGT → mRNA AUG GCA → Met–Ala.

**Mutations:** substitution (point; may change one amino acid, none, or create a stop codon); insertion or deletion (shifts the reading frame: frameshift).

## Mendelian Genetics

**Allele:** version of a gene. **Genotype** (AA, Aa, aa) vs **phenotype** (observed trait). Homozygous = two identical alleles; heterozygous = two different. **Law of segregation:** allele pairs separate into gametes. **Independent assortment:** genes on different chromosomes sort independently.

**EXAMPLE Punnett square, Aa × Aa**

| | A | a |
|---|---|---|
| **A** | AA | Aa |
| **a** | Aa | aa |

Genotypes AA : Aa : aa = 1 : 2 : 1; phenotypes dominant : recessive = 3 : 1; P(aa) = 0.5 × 0.5 = 25 %.

- Test cross Aa × aa: P(aa) = 0.5 × 1 = 50 %.
- Dihybrid AaBb × AaBb, out of 16: 9 : 3 : 3 : 1 (e.g., both dominant = 0.75 × 0.75 × 16 = 9).

**Non-Mendelian:** incomplete dominance (blend: red × white → pink); codominance (both expressed: blood type AB); multiple alleles (ABO); X-linked recessive traits (e.g., hemophilia) appear more often in males.

## Macromolecules & Enzymes

| Class | Building block | Examples / role |
|---|---|---|
| **Carbohydrates** | Monosaccharides (e.g. glucose) | Glucose, sucrose, starch, glycogen, cellulose. Quick energy, storage, plant structure |
| **Lipids** | Glycerol + fatty acids (not true polymers) | Triglycerides, phospholipids, steroids (cholesterol). Long- |

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| Class | Building block | Examples / role |
|---|---|---|
| | | term energy, membranes, hormones, insulation |
| **Proteins** | Amino acids joined by peptide bonds | Enzymes, hemoglobin, collagen, antibodies. Catalysis, structure, transport, defense; shape sets function |
| **Nucleic acids** | Nucleotides (sugar + phosphate + base) | DNA, RNA (ATP is a nucleotide). Store and express genetic information |

Dehydration synthesis joins monomers (releases water); hydrolysis splits them (uses water). **Enzymes** are mostly proteins: they lower activation energy, fit specific substrates at the active site and are not used up; heat and extreme pH denature them.

## Microorganisms & Disease

| Group | Structure | Examples / treatment |
|---|---|---|
| **Bacteria** | Prokaryotic cells; peptidoglycan wall; cocci, bacilli, spirilla | Strep throat, tuberculosis; treated with antibiotics |
| **Viruses** | Not cells: DNA or RNA in a protein capsid; replicate only inside host cells | Influenza, HIV; antibiotics do not work; antivirals, vaccines |
| **Fungi** | Eukaryotic; chitin cell wall; yeasts and molds | Candidiasis, athlete's foot; antifungals |
| **Protozoa** | Single-celled eukaryotes | Malaria (Plasmodium), giardiasis |

**Gram stain:** Gram-positive = thick peptidoglycan wall, stains purple; Gram-negative = thin wall plus outer membrane, stains pink.

**Pathogens** cause disease by invading tissue or releasing toxins; normal flora (e.g., gut bacteria) protect the host and help digestion. Hand hygiene breaks the chain of infection between reservoir and susceptible host.

| Transmission | How | Example |
|---|---|---|
| **Direct contact** | Touching, kissing, sexual contact, contact with soil or vegetation | — |
| **Droplet spread** | Large respiratory droplets sprayed over a short distance (sneezing, coughing, talking) | Pertussis, influenza |
| **Airborne** | Small particles that stay suspended in air, carried by dust or air currents | Pulmonary tuberculosis |
| **Vehicle-borne** | Contaminated food, water, blood products or objects (fomites) | Hepatitis B via blood |
| **Vector-borne** | Carried by a living vector, usually an insect | Malaria (mosquito) |

# Chemistry

8 scored items.

## Atomic Structure & Periodic Table

| Particle | Charge | Location |
|---|---|---|
| **Proton** | positive | nucleus; their number is the atomic number $Z$ and defines the element |
| **Neutron** | none | nucleus |
| **Electron** | negative | shells (energy levels) around the nucleus; valence electrons drive bonding |

> Mass number $A = Z + N$; **isotopes** = same $Z$, different $N$; **ion** = atom that gained (anion, −) or lost (cation, +) electrons. *E.g.* carbon-14: $N = 14 - 6 = 8$; chlorine-35: $N = 35 - 17 = 18$; $Na^+$: 11 protons − 10 electrons = charge +1.

**Periodic table:** rows (periods) = number of electron shells; columns (groups) = same valence electrons, similar properties. Metals on the left, nonmetals on the right, metalloids along the staircase; alkali metals (far-left column) are very reactive; noble gases (far-right column) are stable and unreactive. Electronegativity increases toward the upper right (fluorine highest); atomic radius increases toward the lower left.

## Chemical Bonds

| Bond | How | Example |
|---|---|---|
| **Ionic** | Electrons transferred (metal → nonmetal); oppositely charged ions attract | NaCl |
| **Nonpolar covalent** | Electrons shared equally | $H_2$, $O_2$ |

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| Bond | How | Example |
|---|---|---|
| **Polar covalent** | Electrons shared unequally: partial charges (δ+ / δ−) | $H_2O$ |
| **Hydrogen bond** | Weak attraction between H bonded to N/O/F and a nearby N/O/F | Between water molecules, DNA base pairs |
| **Metallic** | Cations in a sea of shared electrons | Cu, Fe |

## Properties & States of Matter

**Solid:** fixed shape and volume. **Liquid:** fixed volume, takes the container's shape. **Gas:** neither; compressible.

| Phase change | From | To | Heat |
|---|---|---|---|
| **Melting** | solid | liquid | absorbed |
| **Freezing** | liquid | solid | released |
| **Vaporization (evaporation / boiling)** | liquid | gas | absorbed |
| **Condensation** | gas | liquid | released |
| **Sublimation** | solid | gas | absorbed |
| **Deposition** | gas | solid | released |

**Density** = $m/V$ (a physical property): 50 g occupying 20 mL = 2.5 g/mL; a less dense object floats on a denser liquid.

**Physical change:** same substance (melting, dissolving, cutting). **Chemical change:** new substance (signs: gas, color change, precipitate, heat or light). Water melts at 0 °C and boils at 100 °C (1 atm); ice floats because hydrogen bonding makes it less dense than liquid water; water has a high specific heat, cohesion and adhesion.

### Gas laws (temperature in kelvins)

$$P_1V_1 = P_2V_2 \qquad \frac{V_1}{T_1} = \frac{V_2}{T_2} \qquad \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Boyle (constant $T$): pressure up, volume down; this is how breathing moves air. Charles (constant $P$): heat a gas, it expands. Gay-Lussac (constant $V$): heat a gas, its pressure rises. *E.g.* 6 L at 1 atm squeezed to 2 atm: $V_2 = 1 \times 6/2 = 3$ L.

## Chemical Reactions

| Type | Pattern | Example |
|---|---|---|
| **Synthesis** | A + B → AB | $2\ H_2 + O_2 \rightarrow 2\ H_2O$ |
| **Decomposition** | AB → A + B | $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$ |
| **Single replacement** | A + BC → AC + B | $Zn + 2\ HCl \rightarrow ZnCl_2 + H_2$ |
| **Double replacement** | AB + CD → AD + CB | $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ |
| **Combustion** | fuel + $O_2$ → $CO_2$ + $H_2O$ | $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ |
| **Neutralization** | acid + base → salt + water | $HCl + NaOH \rightarrow NaCl + H_2O$ |

### Balancing: conservation of mass

Change coefficients only, never subscripts. $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$: O left 2 × 2 = 4, O right 2 + 2 = 4; H right 2 × 2 = 4, as in $CH_4$.

**Faster reactions:** higher temperature, higher concentration, more surface area, a catalyst (lowers activation energy, not consumed). **Exothermic** releases heat; **endothermic** absorbs it. **Le Chatelier:** a system at equilibrium shifts to offset a change in concentration, temperature or pressure.

## Solutions

Solute dissolves in solvent (water: the "universal solvent"); like dissolves like (polar with polar). Unsaturated → saturated → supersaturated. Most solids dissolve better when hot; gases dissolve better when cold and under pressure.

### Concentration

$$M = \frac{\text{mol solute}}{\text{L solution}} \qquad C_1V_1 = C_2V_2 \qquad \%\ (m/v) = \frac{\text{g solute}}{100\ \text{mL solution}}$$

- NaCl molar mass 22.99 + 35.45 = 58.44 g/mol; 14.61 g in 0.5 L: $M$ = 14.61/(58.44 × 0.5) = 0.50 M.
- 500 mL of 0.5 M from 2 M stock: $V_1$ = 0.5 × 500/2 = 125 mL, then add solvent to 500 mL.
- 250 mL of 5 % (m/v) glucose holds 5 × 250/100 = 12.5 g.
- Moles $n = m/M$: $H_2O$ = 2 × 1.008 + 15.999 ≈ 18.0 g/mol; 36.03 g of water = 2.0 mol.

## Acids & Bases

**Acid:** donates $H^+$, pH < 7, sour, turns blue litmus red. **Base:** accepts $H^+$ (or releases $OH^-$), pH > 7, bitter, slippery, turns red litmus blue. Neutral = 7 (25 °C); scale 0–14.

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$$pH = -\log[H^+] \quad pH + pOH = 14 \quad (25\ °C)$$

Each pH unit is a tenfold change: pH 3 has $10^{6-3}$ = 1,000 times the $[H^+]$ of pH 6. At pH 3, pOH = 14 − 3 = 11.

Strong acids (HCl) ionize fully; weak acids (acetic, carbonic) only partly. **Buffers** resist pH change: the bicarbonate buffer keeps blood pH at 7.35–7.45.

# Scientific Reasoning

9 scored items.

## Scientific Method

Observation → question → **hypothesis** (testable, falsifiable prediction) → controlled experiment → data analysis → conclusion → repeat and communicate.

| Term | Meaning |
|---|---|
| **Independent variable** | What the investigator changes (x-axis) |
| **Dependent variable** | What is measured in response (y-axis) |
| **Controlled variables** | Kept the same in every trial |
| **Control group** | Gets no treatment: the baseline for comparison |
| **Theory** | Broad, well-tested explanation |
| **Law** | Describes what happens (often an equation), not why |

**Accuracy** = closeness to the true value; **precision** = closeness of repeated results to each other. More trials and larger samples make results more reliable. Correlation does not prove causation. An **observation** is what you sense or measure; an **inference** is a conclusion drawn from it. Inductive reasoning goes from specific cases to a general rule; deductive reasoning applies a general rule to a specific case.

### Working with data

- Mean of trials 12, 15, 18: (12 + 15 + 18)/3 = 15.
- Slope: 4 cm on day 2, 10 cm on day 8: (10 − 4)/(8 − 2) = 1 cm/day.
- Percent error: measured 10.4, accepted 10: |10.4 − 10|/10 × 100 = 4 %.

### Judging an investigation

- Only one independent variable should change; everything else stays controlled.
- Look for a control group, enough subjects and repeated trials.
- A conclusion must stay within the data: no claims beyond the conditions tested.
- A hypothesis is supported or not supported, never “proven”.
- Check graphs for title, labeled axes, units and scale before reading a trend.

**Graphs:** line = change over time or a continuous variable; bar = compare categories; pie = parts of a whole; scatter = relationship between two variables (positive, negative or none). Interpolation reads between data points; extrapolation goes beyond them (less reliable).

## Measurement & Units

**Tools:** balance (mass), graduated cylinder (volume: read the bottom of the meniscus at eye level), ruler or meter stick (length), thermometer (temperature), pipette (small volumes). **Mass** (kg, same everywhere) is not **weight** (a force that depends on gravity).

| Prefix | Symbol | × base unit |
|---|---|---|
| **kilo** | k | 1,000 |
| **(base unit)** | — | — |
| **deci** | d | 0.1 |
| **centi** | c | 0.01 |
| **milli** | m | 0.001 |
| **micro** | µ | 0.000001 |

1 in = 2.54 cm · 1 ft = 12 in · 1 lb ≈ 0.4536 kg · 1 kg ≈ 2.2 lb · 1 L = 1,000 mL · 1 mL = 1 $cm^3$

### Conversions

- 2.5 L = 2,500 mL · 250 mg = 0.25 g · 0.5 mg = 500 µg · 0.75 km = 750 m
- 150 lb × 0.45359237 ≈ 68.0 kg
- 5 ft 6 in = (5 × 12 + 6) in × 2.54 ≈ 167.6 cm
- °C = (°F − 32)/1.8: 98.6 °F = 37 °C; °F = 1.8 × °C + 32: 40 °C = 104 °F
- K = °C + 273.15: 37 °C = 310.15 K

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Moving to a smaller unit gives a bigger number (multiply); moving to a larger unit gives a smaller number (divide).

## Common Traps

**TRAP**

- Pulmonary arteries carry deoxygenated blood and pulmonary veins oxygenated blood: arteries are defined by direction (away from the heart), not by $O_2$.
- Tendons join muscle to bone; ligaments join bone to bone.
- Insulin lowers blood glucose; glucagon raises it. Calcitonin lowers blood $Ca^{2+}$; PTH raises it.
- Mitosis gives two genetically identical cells (diploid); meiosis gives four genetically different cells (haploid).
- RNA has uracil, not thymine; codons are read on mRNA.
- Aa × Aa gives a 3 : 1 phenotype ratio but a 1 : 2 : 1 genotype ratio.
- Antibiotics act on bacteria, not on viruses.
- Lower pH = more acidic; one pH unit = tenfold change in $[H^+]$.
- Balance with coefficients, never by changing subscripts.
- The independent variable is changed; the dependent variable is measured.
- Precise is not the same as accurate.
- Supine is face up; prone is face down.
- Universal red-cell donor is O−, universal recipient AB+: Rh counts as well as ABO.
- ADH saves water; aldosterone saves sodium (water follows).
- Food passes through the pharynx but never through the liver or pancreas.
- Gas-law calculations need kelvins, not degrees Celsius.

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